آمار و احتمال مهندسی

Engineering Probability & Statistics

Lecture Notes ▪ جزوه درسی

Module 14

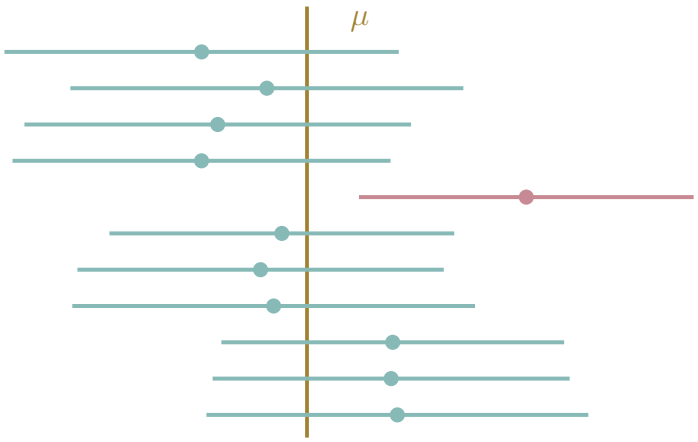
Confidence Intervals

بازه‌های اطمینان

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Learning Objectives

After completing this module, students will be able to:

1. Correctly interpret confidence intervals (frequentist interpretation)
2. Construct CIs for means (known and unknown variance)
3. Apply the t-distribution for small samples
4. Construct CIs for variance
5. Determine required sample sizes
6. Avoid common misconceptions about CIs

14.1 Introduction: Why Point Estimates Aren't Enough

A point estimate (e.g., $\bar{X} = 3.47$) gives no information about **precision**. We need to quantify uncertainty:

"The true mean is estimated to be 3.47, **give or take** some amount."

A **confidence interval** provides a range that, with a specified confidence level, captures the true parameter.

14.2 Correct Interpretation of Confidence Intervals

✔ CORRECT (Frequentist) Interpretation

> "If we repeat this experiment many times, each time computing a 95% CI, then approximately 95% of those intervals will contain the true parameter μ."

The **procedure** has a 95% success rate. Each specific interval either contains μ or it doesn't.

✘ WRONG Interpretation

> ~~ "There is a 95% probability that μ is inside this particular interval $[2.1,\ 4.8]$." ~~

Why this is wrong: μ is a fixed (non-random) number. It's either in $[2.1, 4.8]$ or not — there is no probability about it. The **interval** is random (varies from sample to sample), not the parameter.

Analogy

Think of CIs like a fishing net:

- Before you cast: 95% chance your net will catch the fish (the procedure works 95% of the time)
- After you cast: the fish is either caught or not — no probability anymore
- The fish (μ) doesn't move; your net (CI) does

14.3 CI for Mean — Known Variance (Z-interval)

Setup

Data: $X_1, \ldots, X_n$ i.i.d. from $N(\mu, \sigma^2)$ with σ^2 **known**.

Derivation

$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$, so $Z = \frac{\bar{X}-\mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0, 1)$

$P(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}) = 1 - \alpha$

Solving for μ:

$$P\left(\bar{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

Formula

CI: $\bar{X} \pm z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$

Common z-values

Confidence Level	α	$z_{\alpha/2}$
90%	0.10	1.645
95%	0.05	1.960
99%	0.01	2.576

Example

Noise measurements: $n = 36$, $\bar{X} = 0.52\,\text{V}$, $\sigma = 0.12\,\text{V}$ (known from equipment spec).
95% CI: $0.52 \pm 1.96 \times \frac{0.12}{\sqrt{36}} = 0.52 \pm 0.0392 = **[0.481,\ 0.559]V**$

14.4 CI for Mean — Unknown Variance (t-interval)

The Problem

Usually σ is unknown. We estimate it with S. But $\frac{\bar{X}}{\frac{S}{\sqrt{n}}}$ does NOT follow $N(0,\,1)$.

Student's t-Distribution

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t(n-1)$$

The t-distribution:

- Symmetric about 0
- Heavier tails than $N(0,\,1)$ (more uncertainty because S is estimated)
- Degrees of freedom: $\nu = n - 1$
- As $\nu \to \infty$: $t(\nu) \to N(0,\,1)$
- For $\nu \geq 30$: $t \approx z$ (practically)

Formula

CI: $\bar{X} \pm t_{\alpha/2,n-1} \cdot \dfrac{S}{\sqrt{n}}$

Example

Component delay: $n = 15$, $\bar{X} = 2.34$ ns, $S = 0.45$ ns.
95% CI: $t_{0.025,14} = 2.145$
CI: $2.34 \pm 2.145 \times \frac{0.45}{\sqrt{15}} = 2.34 \pm 0.249 = **[2.09,\,2.59]$ ns**
(Wider than z-interval because of additional uncertainty in S.)

14.5 CI for Variance

Setup

Data from $N(\mu,\, \sigma^2)$. Want CI for σ^2.

Key Distribution

$$(n-1)S^2/\sigma^2 \sim \chi^2(n-1)$$

Formula

$$\left[\frac{(n-1)S^2}{\chi^2_{\alpha/2,n-1}},\quad \frac{(n-1)S^2}{\chi^2_{1-\alpha/2,n-1}}\right]$$

Example

Noise power: $n = 20$, $S^2 = 0.045\,\mathrm{V}^2$.
95% CI for σ^2: $\chi^2_{0.025,19} = 32.852$, $\chi^2_{0.975,19} = 8.907$
CI: $\left[\frac{(19)(0.045)}{32.852},\, \frac{(19)(0.045)}{8.907}\right] = [0.026,\, 0.096]V^2$
95% CI for σ: $\left[\sqrt{0.026},\, \sqrt{0.096}\right] = [0.161,\, 0.310]V$

14.6 CI for Proportions

Setup

n trials, $\hat{x} = \frac{k}{n}$ successes. Estimate p = true proportion.

Large-Sample Formula (Wald Interval)

$$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Example: Bit Error Rate

Transmitted 100,000 bits, observed 53 errors. $\hat{p} = \frac{53}{100000} = 5.3 \times 10^{-4}$.

95% CI: $5.3 \times 10^{-4} \pm 1.96\sqrt{5.3 \times 10^{-4} \times \frac{0.9995}{100000}} = 5.3 \times 10^{-4} \pm 1.43 \times 10^{-4}$

CI: $[3.9 \times 10^{-4},\ 6.7 \times 10^{-4}]$

14.7 Factors Affecting CI Width

Factor	Effect on Width	Explanation
↑ Sample size n	Narrower	More data → more precision
↑ Confidence level	Wider	More confidence → wider net
↑ Variability σ	Wider	More noise → harder to pin down μ

The Tradeoff

- Want narrow CI (precise) → need more data or accept lower confidence
- Want high confidence → accept wider CI or collect more data
- Want both narrow AND high confidence → need LOTS of data

14.8 Sample Size Determination

Question: How many measurements *do* I need?

For CI width $\leq 2E$ (margin of error E):

$$n \geq \left(\frac{z_{\alpha/2} \cdot \sigma}{E}\right)^2$$

Example

Want to estimate mean delay within ± 0.5 ms with 95% confidence. Expect $\sigma \approx 3$ ms.
$n \geq \left(1.96 \times \frac{3}{0.5}\right)^2 = (11.76)^2 = 138.3 \rightarrow$ need $n = 139$ **measurements**

14.9 Practice Problems

Problem 1

A sensor measures voltage. From 50 measurements: $\bar{X} = 3.32\,\text{V}$, $S = 0.15\,\text{V}$.

(a) Construct 95% CI for μ. (b) Construct 99% CI for μ. (c) Which is wider and why?

Solution

(a) $t_{0.025,49} \approx 2.010$. CI: $3.32 \pm 2.010\left(\frac{0.15}{\sqrt{50}}\right) = 3.32 \pm 0.043 = [3.277,\ 3.363]$

(b) $t_{0.005,49} \approx 2.680$. CI: $3.32 \pm 2.680\left(\frac{0.15}{\sqrt{50}}\right) = 3.32 \pm 0.057 = [3.263,\ 3.377]$

(c) 99% CI is wider — higher confidence requires a wider interval.

Problem 2

From 25 noise power measurements: $S^2 = 0.048\,\text{V}^2$. Construct 95% CI for σ^2.

Solution $\chi^2_{0.025,24} = 39.364$, $\chi^2_{0.975,24} = 12.401$

CI: $\left[\frac{24(0.048)}{39.364},\ \frac{24(0.048)}{12.401}\right] = [0.029,\ 0.093]V^2$

Problem 3

How many BER measurements needed to estimate BER within $\pm 20\%$ of its true value with 95% confidence, if expected BER $\approx 10^{-3}$?

Solution For proportion: $n \geq \frac{z^2 p(1-p)}{E^2}$ where $E = 0.2p = 2 \times 10^{-4}$.

$n \geq \frac{(1.96)^2(10^{-3})(0.999)}{(2\times10^{-4})^2} = 3.84 \times \frac{10^{-3}}{4\times10^{-8}} \approx 96{,}000$ bits.

Need to transmit approximately 96,000 bits (or observe ~ 96 errors).

Next Module: Module 15 — Hypothesis Testing